

FREDERICTON AQUATIC CENTRE, NB, Canada

WMO: 716445

Lat: 45.966N

Lon: 66.651W

Elev: 27

StdP: 14.68

Time Zone: -4.00 (NAA)

Period: 05-15

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -8.6	(c) -2.9	(d) -15.8	(e) 2.3	(f) -6.3	(g) -11.1	(h) 3.0	(i) -1.9	(j) 27.7	(k) 16.9	(l) 25.6	(m) 16.2	(n) 5.4	(o) 280	(p) 0.515

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	17.8	86.7	71.4	83.7	69.9	81.1	67.9	74.1	82.7	72.5	80.1	71.0	77.5	7.9	280

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
71.4	116.1	77.8	70.0	110.5	76.4	68.6	105.1	75.0	37.6	83.5	36.2	80.6	34.8	77.4	80.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 23.5	(b) 20.5	(c) 17.9	(e) -15.2	(f) 92.3	(g) 7.9	(h) 2.3	(i) -20.9	(j) 93.9	(k) -25.5	(l) 95.2	(m) -30.0	(n) 96.5	(o) -35.7	(p) 98.2		
(4) 23.5	(5) 20.5	(6) 17.9	(7) -15.6	(8) 77.2	(9) 7.6	(10) 1.9	(11) -21.1	(12) 78.5	(13) -25.6	(14) 79.6	(15) -29.9	(16) 80.7	(17) -35.4	(18) 82.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	45.9	19.2	19.7	30.2	43.3	54.7	63.9	70.9	69.2	61.9	50.2	39.2	26.7
	DBStd	19.78	12.46	10.69	10.01	6.71	6.74	5.58	4.55	4.93	6.71	7.12	8.32	9.86
	HDD50	3810	954	848	617	219	25	0	0	0	1	86	338	722
	HDD65	7413	1419	1268	1079	652	328	84	7	17	137	461	774	1187
	CDD50	2318	0	0	2	18	172	419	649	595	358	92	13	0
	CDD65	446	0	0	0	0	9	52	191	147	45	2	0	0
	CDH74	3154	0	0	2	10	140	423	1334	950	279	16	0	0
	CDH80	796	0	0	0	1	36	93	360	232	73	1	0	0
	WSAvg	7.0	8.0	8.3	8.5	7.6	6.6	6.0	5.3	5.3	5.8	7.0	7.0	8.1
	PrecAvg	44.9	3.4	2.9	3.6	3.3	3.7	3.7	3.9	3.2	3.9	4.5	4.3	4.4
(15) Precipitation	PrecMax	59.0	6.0	5.2	5.9	5.8	7.6	7.2	8.0	7.2	9.0	10.2	7.0	7.6
	PrecMin	28.4	1.9	1.1	1.3	1.1	0.9	1.7	1.1	0.7	0.8	1.3	1.8	1.6
	PrecStd	6.1	1.1	1.0	1.2	1.3	1.5	1.4	2.0	1.7	2.1	2.3	1.3	1.6
	PrecStd	6.1	1.1	1.0	1.2	1.3	1.5	1.4	2.0	1.7	2.1	2.3	1.3	1.6
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	53.2	47.8	63.6	72.2	84.6	86.9	90.5	89.4	88.1	76.3	65.3	55.7
		MCWB	52.0	44.3	51.9	56.1	67.5	68.6	73.5	73.8	73.9	64.3	62.8	53.4
	2%	DB	46.0	40.9	52.0	65.3	77.6	82.7	86.9	85.7	80.8	68.8	60.5	48.4
		MCWB	44.1	36.6	44.1	51.5	61.1	68.0	71.7	71.8	69.1	60.3	56.5	46.9
	5%	DB	40.0	37.4	47.3	59.9	72.4	78.9	83.9	82.3	76.4	64.8	56.7	43.9
		MCWB	37.6	34.7	40.4	49.3	58.5	65.6	70.6	69.3	67.1	59.8	53.3	42.0
	10%	DB	36.3	35.3	43.4	55.3	67.5	75.3	81.3	79.4	73.1	61.4	52.3	40.3
		MCWB	34.5	33.5	38.0	46.3	56.1	63.3	68.7	68.2	65.1	56.2	49.1	38.2
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	52.7	45.5	54.7	60.9	69.4	72.9	76.5	75.8	74.6	67.4	63.3	53.4
		MCDB	53.7	47.4	62.6	69.1	80.0	82.1	86.4	84.7	85.5	72.1	65.1	55.3
	2%	WB	44.3	37.2	45.9	55.2	64.5	70.6	74.3	73.8	71.8	63.5	58.1	46.8
		MCDB	45.6	39.3	50.1	60.9	72.7	78.4	82.9	82.3	77.0	66.7	59.4	47.7
	5%	WB	37.8	35.4	41.5	50.8	61.4	68.0	72.7	72.2	69.4	60.9	53.9	42.4
		MCDB	39.5	37.0	46.1	57.0	68.8	74.2	80.4	79.4	73.9	65.6	56.3	43.4
	10%	WB	34.6	33.7	38.4	47.4	58.4	65.8	71.1	70.4	66.8	57.2	49.1	38.3
		MCDB	36.2	35.3	42.5	54.0	64.7	72.1	78.0	76.2	72.0	59.8	51.5	39.7
(35) Mean Daily Temperature Range	5% DB	MDBR	17.3	17.9	18.0	17.7	18.6	17.6	17.8	17.5	18.0	14.6	13.5	13.1
		MCDBR	19.4	17.2	22.6	27.8	28.2	24.7	22.0	21.9	21.4	20.0	20.0	16.5
		MCWBR	18.7	14.0	15.4	16.1	14.2	11.6	9.9	10.0	11.6	13.4	16.5	16.1
	5% WB	MCDBR	20.0	15.0	21.0	23.2	23.0	20.6	18.7	18.7	18.1	17.6	17.9	17.0
		MCWBR	19.2	12.9	16.6	16.6	13.9	11.5	9.6	9.6	11.2	14.0	16.9	16.9
(40) Clear-Sky Solar Irradiance	taub		0.275	0.286	0.318	0.346	0.374	0.406	0.425	0.400	0.371	0.328	0.306	0.287
	taud		2.455	2.401	2.425	2.424	2.388	2.305	2.272	2.344	2.400	2.525	2.538	2.475
	Ebn at Noon		267	285	290	290	283	273	266	269	267	267	253	249
	Edh at Noon		21	27	31	34	37	40	41	37	32	24	20	19
(44) All-Sky Solar Radiation	RadAvg		473	749	1113	1381	1584	1713	1739	1538	1242	766	449	349
	RadStd		50	68	74	130	180	129	138	98	92	72	52	32

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (32 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+90	N/A

Nomenclature: See separate page